Customer Retention & Churn Analysis

1. Introduction

This project focuses on analyzing customer churn behavior using the Telco Customer Churn dataset. The objective is to identify patterns in customer attrition, understand retention drivers, and provide actionable recommendations to improve customer retention.

2. Data Cleaning & Preparation

The dataset was cleaned by handling missing values, correcting data types, and ensuring consistency. The TotalCharges column was converted to numeric format, and invalid records were removed. A new column 'Churn_Flag' was created to represent churn status numerically for analysis.

3. Data Analysis

3.1 Churn Distribution

Out of 7032 customers, 1869 customers have churned, resulting in a churn rate of approximately 27%. This indicates a significant portion of customers leaving the service.

3.2 Churn by Contract Type

Customers with month-to-month contracts exhibit the highest churn rate (~42%), while those with long-term contracts (one-year and two-year) show significantly lower churn rates.

3.3 Churn by Payment Method

Customers using electronic check have the highest churn rate (~45%), whereas customers using automatic payment methods such as credit card or bank transfer show lower churn rates.

3.4 Churn by Internet Service

Fiber optic users show higher churn (~41%) compared to DSL users (~19%) and customers without internet service.

3.5 Churn by Tenure

Customers with shorter tenure (0–1 year) have the highest churn rate (~47%), while long-term customers (4+ years) show very low churn (~9%), indicating strong retention over time.

4. Dashboard Overview

A Power BI dashboard was developed to visualize key metrics such as total customers, churn rate, average tenure, and churned customers. The dashboard includes segmentation by contract type, payment method, internet service, and tenure groups.

5. Key Insights

- Month-to-month contracts are the primary driver of churn.
- New customers are at higher risk of leaving.
- Electronic check users are more likely to churn.
- Fiber optic customers show higher churn behavior.
- Long-term customers demonstrate strong loyalty.

6. Recommendations

- Encourage customers to switch to long-term contracts through discounts.
- Improve onboarding experience for new customers.
- Promote automatic payment methods.
- Investigate service quality issues for fiber optic users.
- Implement targeted retention campaigns for high-risk segments.

7. Conclusion

This analysis highlights key churn drivers and provides actionable strategies to improve customer retention. By addressing high-risk segments and enhancing customer experience, businesses can significantly reduce churn and improve long-term growth.